pset5

May 1, 2026

1 PSET 5

1) implied volatility

[7]:
```
from IPython.display import Image, display
display(Image("1a.png", width=600))
```

① implied volatility

a) for each call option, compute mid price = $\frac{\text{bid} + \text{ask}}{2}$

↳ use Black scholes to compute theoretical option price

$C = S_0 N(d_1) - Ke^{-rT} N(d_2)$

for each strike K, find σ that makes C = mid

given: $S_0 = 6816.89$, $T = 112/365$, $r = 0.0369$

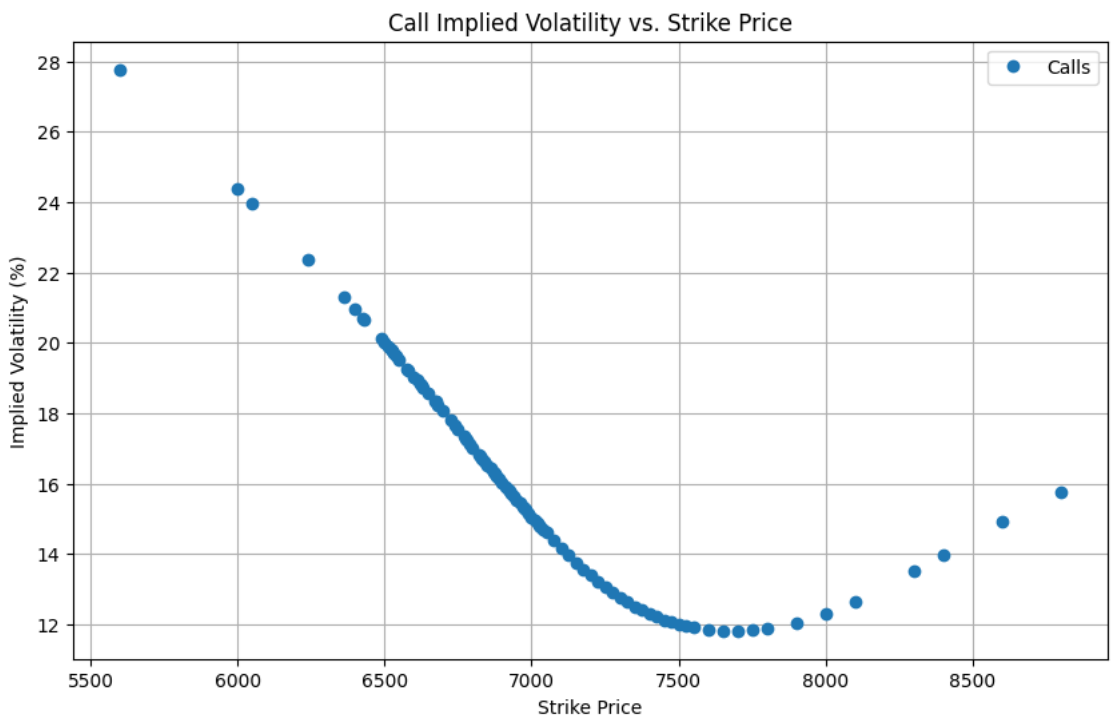
[3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from math import log, sqrt, exp, erf

S0 = 6816.89
T = 112 / 365
r = 0.0369

def N(x):
    return 0.5 * (1 + erf(x / sqrt(2)))

def bs_call(S, K, T, r, sigma):
```

```
    d1 = (log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * sqrt(T))
    d2 = d1 - sigma * sqrt(T)
    return S * N(d1) - K * exp(-r * T) * N(d2)

def bs_put(S, K, T, r, sigma):
    d1 = (log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * sqrt(T))
    d2 = d1 - sigma * sqrt(T)
    return K * exp(-r * T) * N(-d2) - S * N(-d1)

def implied_vol(price, S, K, T, r, option_type, tol=1e-8, max_iter=200):
    low = 1e-8
    high = 5.0

    if option_type == "call":
        price_func = bs_call
    elif option_type == "put":
        price_func = bs_put
    else:
        raise ValueError("option_type must be 'call' or 'put'")

    for _ in range(max_iter):
        mid = (low + high) / 2
        model_price = price_func(S, K, T, r, mid)

        if abs(model_price - price) < tol:
            return mid

        if model_price < price:
            low = mid
        else:
            high = mid

    return (low + high) / 2
```

[4]:
```
calls = pd.read_csv("PSET5 Calls.csv")

calls["mid"] = (calls["bid"] + calls["ask"]) / 2

calls["implied_vol"] = calls.apply(
    lambda row: implied_vol(
        price=row["mid"],
        S=S0,
        K=row["strike"],
        T=T,
        r=r,
        option_type="call"
    ),
```

```
    axis=1
)

display(calls.head())

plt.figure(figsize=(10, 6))
plt.plot(calls["strike"], 100 * calls["implied_vol"], marker="o", linestyle="",␣
 ↪label="Calls")

plt.xlabel("Strike Price")
plt.ylabel("Implied Volatility (%)")
plt.title("Call Implied Volatility vs. Strike Price")
plt.legend()
plt.grid(True)
plt.show()
```

```
   strike     bid     ask      mid  implied_vol
0  5600.0  1314.3  1322.4  1318.35     0.277693
1  6000.0   948.4   955.8   952.10     0.243880
2  6050.0   904.0   911.3   907.65     0.239627
3  6240.0   739.2   746.1   742.65     0.223555
4  6360.0   638.5   645.1   641.80     0.213005
```



[8]:
```
display(Image("1b.png", width=600))
```

b) for each put option, compute mid price $= \frac{\text{bid} + \text{ask}}{2}$

↳ use Black Scholes to compute theoretical option price

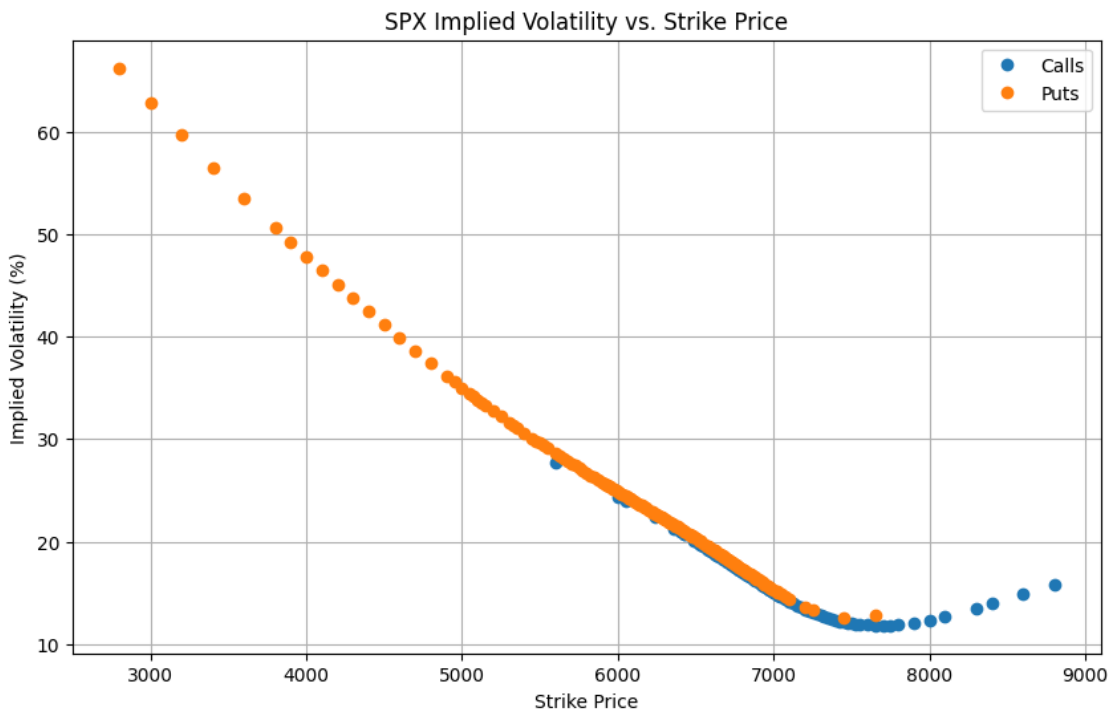
$P = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$

for each strike K, find σ that makes $P = \text{mid}$

given: $S_0 = 6816.89$, $T = 112/365$, $r = 0.0369$

[9]:
```
puts = pd.read_csv("PSET5 Puts.csv")

puts["mid"] = (puts["bid"] + puts["ask"]) / 2

puts["implied_vol"] = puts.apply(
    lambda row: implied_vol(
        price=row["mid"],
        S=S0,
        K=row["strike"],
        T=T,
        r=r,
        option_type="put"
    ),
    axis=1
)

display(puts.head())

plt.figure(figsize=(10, 6))

plt.plot(calls["strike"], 100 * calls["implied_vol"], marker="o", linestyle="",␣
 ↪label="Calls")
plt.plot(puts["strike"], 100 * puts["implied_vol"], marker="o", linestyle="",␣
 ↪label="Puts")

plt.xlabel("Strike Price")
plt.ylabel("Implied Volatility (%)")
plt.title("SPX Implied Volatility vs. Strike Price")
plt.legend()
plt.grid(True)
plt.show()
```

```
   strike  bid  ask   mid  implied_vol
0  2800.0  3.3  3.9  3.60     0.662188
1  3000.0  4.1  4.5  4.30     0.627904
2  3200.0  4.9  5.5  5.20     0.596595
3  3400.0  5.8  6.5  6.15     0.565319
4  3600.0  6.9  7.6  7.25     0.535289
```



[10]:
```
display(Image("1c.png", width=600))
```

c) The resulting graph has the "smile" shape discussed in lectures. Implied volatility is higher for small K, esp. for out-of-the-money puts).

In regions where both puts & calls are available for the same strike, the implied volatility should nearly coincide.

For European options, $C - P = S_0 - Ke^{-rT}$

The Black-Scholes call/put formulas also satisfy this using the same σ $\Rightarrow$ if market put/call prices $\approx$ satisfy put-call parity, the implied volatility backed out from a call & the implied volatility backed out from a put w/ the same K should be nearly the same.

2. delta hedging

[12]:
```
display(Image("2a.png", width=600))
```

②

European call option on SPY w/ strike $K = 600$, $r = 0.0369$

for each date, time T to maturity $= \dfrac{\#\text{ days until } 3/21/25}{365}$

$\rightarrow \quad C = SN(d_1) - Ke^{-rT}N(d_2)$

$\rightarrow \quad \Delta = N(d_1) = \#$ shares of SPY to hold to replicate 1 call

a) on 12/2/2024, $C = 23.73$

$S = 603.67$

$T = \dfrac{109}{365} = 0.2986$

using bisection, $\sigma = 0.13616$

so implied volatility $= 13.616\%$

$\hookrightarrow$ so, $\Delta = N(d_1) = \boxed{0.60502}$ to replicate 1 call option

[13]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from math import log, sqrt, exp, erf

K = 600
r = 0.0369
maturity = pd.Timestamp("2025-03-21")

def N(x):
    return 0.5 * (1 + erf(x / sqrt(2)))

def bs_call(S, K, T, r, sigma):
    d1 = (log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * sqrt(T))
    d2 = d1 - sigma * sqrt(T)
    return S * N(d1) - K * exp(-r * T) * N(d2)

def call_delta(S, K, T, r, sigma):
    d1 = (log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * sqrt(T))
    return N(d1)
```

```python
def implied_vol_call(price, S, K, T, r, tol=1e-8, max_iter=200):
    low = 1e-8
    high = 5.0

    for _ in range(max_iter):
        mid = (low + high) / 2
        model_price = bs_call(S, K, T, r, mid)

        if abs(model_price - price) < tol:
            return mid

        if model_price < price:
            low = mid
        else:
            high = mid

    return (low + high) / 2

hedge = pd.read_csv("hedge.csv")
hedge["Date"] = pd.to_datetime(hedge["Date"], format="%m/%d/%Y")

# First day: December 2, 2024
S0 = hedge.loc[0, "spot"]
C0 = hedge.loc[0, "option_price"]
date0 = hedge.loc[0, "Date"]

T0 = (maturity - date0).days / 365

sigma0 = implied_vol_call(C0, S0, K, T0, r)
delta0 = call_delta(S0, K, T0, r, sigma0)

print("Part (a)")
print(f"Date: {date0.date()}")
print(f"SPY price: {S0:.2f}")
print(f"Option price: {C0:.2f}")
print(f"T: {T0:.6f}")
print(f"Implied volatility: {sigma0:.6f}")
print(f"Implied volatility (%): {100 * sigma0:.2f}%")
print(f"Delta: {delta0:.6f}")
print(f"Shares of SPY to hold: {delta0:.6f}")
```

```
Part (a)
Date: 2024-12-02
SPY price: 603.63
Option price: 23.33
T: 0.298630
Implied volatility: 0.136163
```

```
Implied volatility (%): 13.62%
Delta: 0.605019
Shares of SPY to hold: 0.605019
```

[15]:
```python
display(Image("2b.png", width=600))
```

b)

amount invested in risk-free asset = initial capital − ΔS

$= 23.33 - (0.60502)(603.67) = -341.88$

→ borrow money @ the risk-free rate

on 12/3/24, SPY is 603.91 ⇒ portfolio $V_1 = \Delta S_1 + B_0 e^{r/365}$

$= \boxed{23.46}$

[16]:
```python
# Initial cash position after buying delta shares
cash0 = C0 - delta0 * S0

# Next day: December 3, 2024
S1 = hedge.loc[1, "spot"]
date1 = hedge.loc[1, "Date"]

dt = (date1 - date0).days / 365

portfolio_value_dec3 = delta0 * S1 + cash0 * exp(r * dt)

print("Part (b)")
print(f"Initial cash position: {cash0:.6f}")
print(f"SPY price on Dec 3: {S1:.2f}")
print(f"Portfolio value on Dec 3: {portfolio_value_dec3:.6f}")
print(f"Market option price on Dec 3: {hedge.loc[1, 'option_price']:.6f}")
```

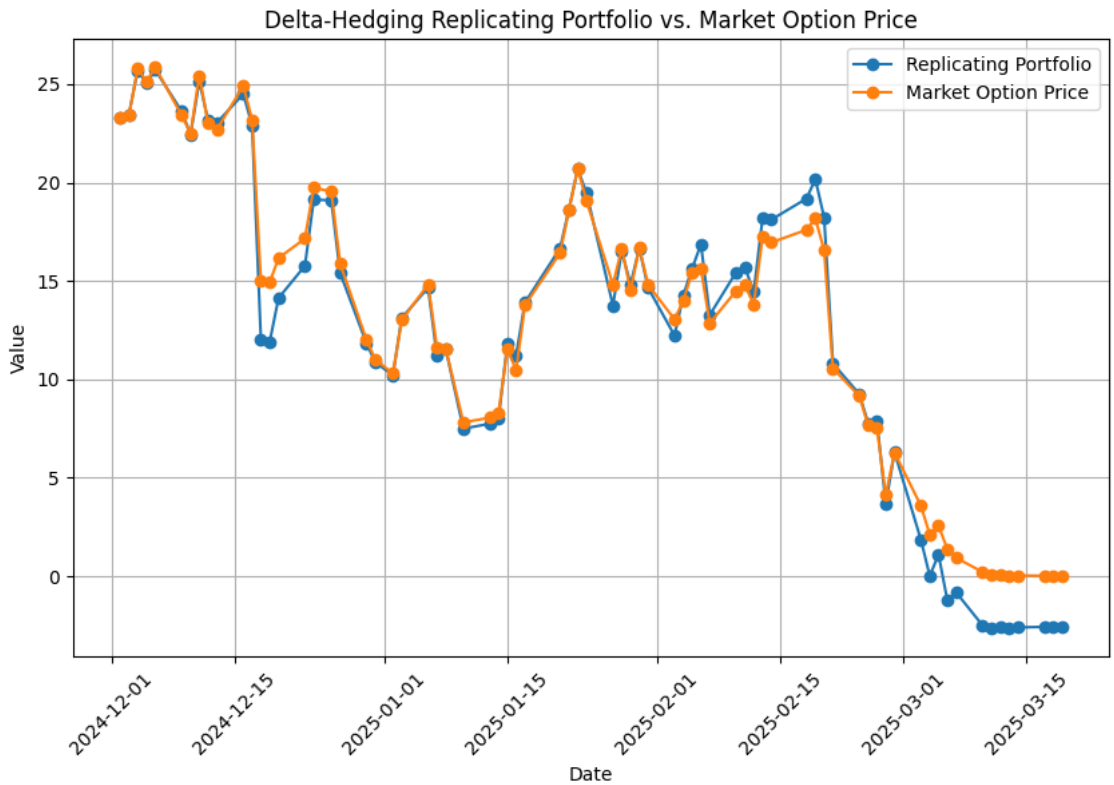
```
Part (b)
Initial cash position: -341.877459
SPY price on Dec 3: 603.91
Portfolio value on Dec 3: 23.464841
Market option price on Dec 3: 23.400000
```

c)

[19]:
```
portfolio_values = []
implied_vols = []
deltas = []
cash_positions = []

# Initialize on the first day
current_date = hedge.loc[0, "Date"]
S = hedge.loc[0, "spot"]
option_price = hedge.loc[0, "option_price"]
T = (maturity - current_date).days / 365

sigma = implied_vol_call(option_price, S, K, T, r)
delta = call_delta(S, K, T, r, sigma)

portfolio_value = option_price
cash = portfolio_value - delta * S

portfolio_values.append(portfolio_value)
implied_vols.append(sigma)
deltas.append(delta)
cash_positions.append(cash)

# Move through each subsequent trading day
for i in range(1, len(hedge)):
    previous_date = hedge.loc[i - 1, "Date"]
    current_date = hedge.loc[i, "Date"]

    S = hedge.loc[i, "spot"]
    option_price = hedge.loc[i, "option_price"]

    # Carry cash forward at the risk-free rate
    dt = (current_date - previous_date).days / 365
    cash = cash * exp(r * dt)

    # Portfolio value before rebalancing
    portfolio_value = delta * S + cash

    # Recompute implied volatility and delta using today's option price
    T = (maturity - current_date).days / 365
    sigma = implied_vol_call(option_price, S, K, T, r)
    new_delta = call_delta(S, K, T, r, sigma)

    # Rebalance: total portfolio value stays the same
    cash = portfolio_value - new_delta * S
    delta = new_delta

    portfolio_values.append(portfolio_value)
```

```
    implied_vols.append(sigma)
    deltas.append(delta)
    cash_positions.append(cash)

hedge["implied_vol"] = implied_vols
hedge["delta"] = deltas
hedge["portfolio_value"] = portfolio_values
hedge["cash_position"] = cash_positions

display(hedge.head())

plt.figure(figsize=(10, 6))

plt.plot(
    hedge["Date"],
    hedge["portfolio_value"],
    marker="o",
    label="Replicating Portfolio"
)

plt.plot(
    hedge["Date"],
    hedge["option_price"],
    marker="o",
    label="Market Option Price"
)

plt.xlabel("Date")
plt.ylabel("Value")
plt.title("Delta-Hedging Replicating Portfolio vs. Market Option Price")
plt.legend()
plt.grid(True)
plt.xticks(rotation=45)
plt.show()
```

```
        Date    spot  option_price  implied_vol     delta  portfolio_value  \
0 2024-12-02  603.63         23.33     0.136163  0.605019        23.330000
1 2024-12-03  603.91         23.40     0.136277  0.607180        23.464841
2 2024-12-04  607.66         25.80     0.137723  0.637759        25.707068
3 2024-12-05  606.66         25.13     0.138368  0.628936        25.032727
4 2024-12-06  607.81         25.86     0.139335  0.637753        25.719959

   cash_position
0    -341.877459
1    -343.217415
2    -361.833501
3    -356.517614
4    -361.912994
```



[20]:
```
display(Image("2d.png", width=600))
```

d)

No, this replicating portfolio does not match the option price exactly. This is bc the Black-Scholes assumes continuous rebalancing, but here we only rebalance once / trading day. The Black Scholes model also assumes constant volatility, but the volatility in the data changed over time.

3. markowitz portfolio analysis

[22]:
```
display(Image("3a.png", width=600))
```

③

a) for each stock, calculate daily simple return

$$r_{i,t} = \frac{s_{i,t} - s_{i,t-1}}{s_{i,t-1}}, \quad \text{sample } \mu_{daily} = E[r_t]$$

covariance $\Sigma_{daily} = Cov(r_t)$

total

$\rightarrow \mu = 252\,\mu_{daily}$

$\Sigma = 252\,\Sigma_{daily}$

minimum-var portfolio solves: $\min \; w^T \Sigma w$

subj. to $\mathbf{1}^T w = 1$

short selling allowed $\Rightarrow$ closed form sol is $w_{min} = \dfrac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^T\Sigma^{-1}\mathbf{1}}$

[21]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Change this filename if your file is named techprices.csv
prices = pd.read_csv("PSET5 Tech Prices.csv")

prices["Date"] = pd.to_datetime(prices["Date"])

tickers = list(prices.columns[1:])
n = len(tickers)

# Daily simple returns
returns = prices[tickers].pct_change().dropna()

# Daily mean and covariance
mu_daily = returns.mean()
Sigma_daily = returns.cov()

# Annualized mean and covariance
mu = 252 * mu_daily
Sigma = 252 * Sigma_daily
```

```python
mu_vec = mu.values
Sigma_mat = Sigma.values
ones = np.ones(n)

# Part (a): minimum-variance portfolio

# w_min = Sigma^{-1} 1 / (1^T Sigma^{-1} 1)
x = np.linalg.solve(Sigma_mat, ones)
w_min = x / (ones @ x)

minvar_return = mu_vec @ w_min
minvar_std = np.sqrt(w_min @ Sigma_mat @ w_min)

minvar_weights = pd.DataFrame({
    "ticker": tickers,
    "weight": w_min
})

print("Part (a): Minimum-variance portfolio")
print(f"Annualized mean return: {minvar_return:.6f}")
print(f"Annualized standard deviation: {minvar_std:.6f}")

display(minvar_weights)
```

```
Part (a): Minimum-variance portfolio
Annualized mean return: 0.105659
Annualized standard deviation: 0.138138
```

	ticker	weight
0	AAPL	0.058189
1	ACN	-0.038993
2	ADBE	-0.028904
3	ADI	0.048836
4	ADSK	-0.055766
..	…	…
62	UI	-0.016910
63	VRSN	0.077568
64	WDAY	0.007124
65	WDC	-0.027506
66	ZBRA	-0.026070

[67 rows x 2 columns]

[23]:
```python
display(Image("3b.png", width=600))
```

b) mean-variance portfolio solves: $\max \quad \mu^T w - \frac{\gamma}{2} w^T \Sigma w$

subj. to $\mathbf{1}^T w = 1$

$\gamma = 7$

first-order condition: $\mu - \gamma \Sigma w - \lambda \mathbf{1} = 0$

$\Rightarrow \quad w = \frac{1}{\gamma} \Sigma^{-1} (\mu - \lambda \mathbf{1})$

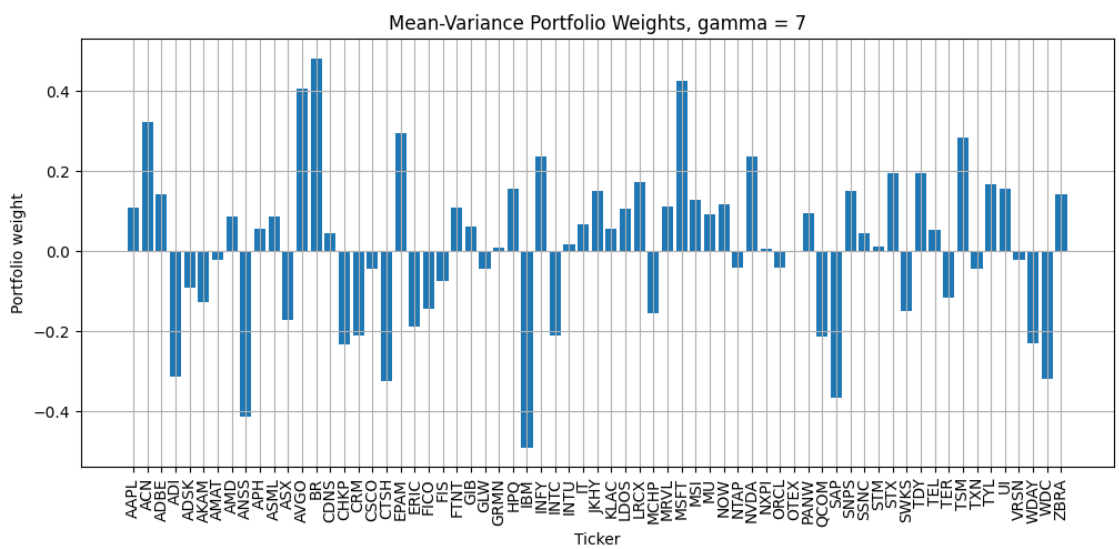
↳ this portfolio is much more aggressive than the min-var portfolio bc it maximizes return, adj by risk adv.

+ short selling is allowed!

[ ]:
```
gamma = 7

# Solve for Sigma^{-1} mu and Sigma^{-1} 1
y = np.linalg.solve(Sigma_mat, mu_vec)
x = np.linalg.solve(Sigma_mat, ones)

A = ones @ x
B = ones @ y

# Lagrange multiplier from budget constraint
lam = (B - gamma) / A

# Mean-variance weights
w_mv = (y - lam * x) / gamma

mv_return = mu_vec @ w_mv
mv_std = np.sqrt(w_mv @ Sigma_mat @ w_mv)

mv_weights = pd.DataFrame({
    "ticker": tickers,
    "weight": w_mv
})

print("Part (b): Mean-variance portfolio")
print(f"Annualized mean return: {mv_return:.6f}")
print(f"Annualized standard deviation: {mv_std:.6f}")
```

```
display(mv_weights)

plt.figure(figsize=(12, 5))
plt.bar(mv_weights["ticker"], mv_weights["weight"])
plt.xlabel("Ticker")
plt.ylabel("Portfolio weight")
plt.title("Mean-Variance Portfolio Weights, gamma = 7")
plt.xticks(rotation=90)
plt.grid(True)
plt.show()
```

```
Part (b): Mean-variance portfolio
Annualized mean return: 1.072596
Annualized standard deviation: 0.396505

   ticker    weight
0    AAPL  0.108095
1     ACN  0.321549
2    ADBE  0.143241
3     ADI -0.312576
4    ADSK -0.091255
..    …       …
62     UI  0.155228
63   VRSN -0.020929
64   WDAY -0.228867
65    WDC -0.318927
66   ZBRA  0.142470

[67 rows x 2 columns]
```



[25]:
```
display(Image("3c.png", width=600))
```

c) minimum variance portfolio problem for target returns m

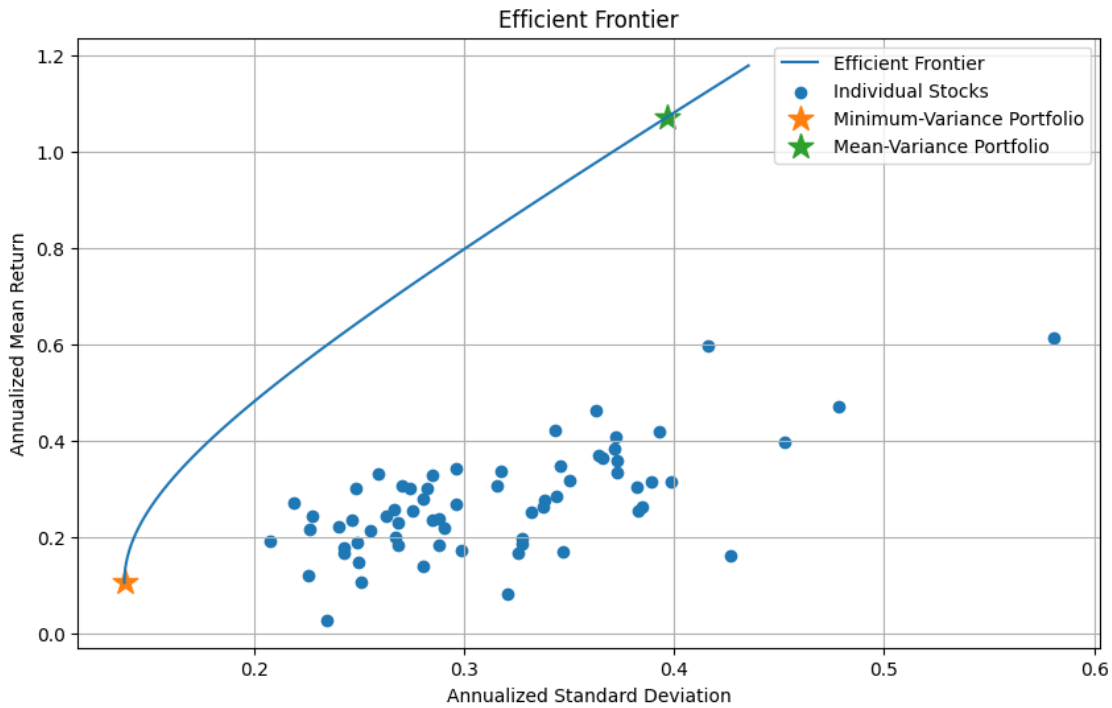
$$\rightarrow \quad \text{minimize} \quad w^T \Sigma w$$
$$\text{subj. to} \quad \mathbf{1}^T w = 1$$
$$\mu^T w = m$$

portfolio st. dev

$$\sigma_p = \sqrt{w^T \Sigma w}$$

[27]:
```
A = ones @ np.linalg.solve(Sigma_mat, ones)
B = ones @ np.linalg.solve(Sigma_mat, mu_vec)
C = mu_vec @ np.linalg.solve(Sigma_mat, mu_vec)
D = A * C - B**2

target_returns = np.linspace(
    minvar_return,
    max(mu.max(), mv_return) * 1.1,
    100
)

frontier_stds = []
frontier_returns = []

for target in target_returns:
    coeff_1 = (C - B * target) / D
    coeff_2 = (A * target - B) / D

    w_target = np.linalg.solve(
        Sigma_mat,
        coeff_1 * ones + coeff_2 * mu_vec
    )

    port_return = mu_vec @ w_target
    port_std = np.sqrt(w_target @ Sigma_mat @ w_target)

    frontier_returns.append(port_return)
    frontier_stds.append(port_std)

# Individual stock points
stock_stds = np.sqrt(np.diag(Sigma_mat))
```

```python
stock_returns = mu_vec

plt.figure(figsize=(10, 6))

plt.plot(frontier_stds, frontier_returns, label="Efficient Frontier")

plt.scatter(stock_stds, stock_returns, label="Individual Stocks")

plt.scatter(
    minvar_std,
    minvar_return,
    marker="*",
    s=200,
    label="Minimum-Variance Portfolio"
)

plt.scatter(
    mv_std,
    mv_return,
    marker="*",
    s=200,
    label="Mean-Variance Portfolio"
)

plt.xlabel("Annualized Standard Deviation")
plt.ylabel("Annualized Mean Return")
plt.title("Efficient Frontier")
plt.legend()
plt.grid(True)
plt.show()
```



[28]:
```
display(Image("3d.png", width=600))
```

d) risk-free asset w/ annualized return $R_f = 0.01$

tangency portfolio is the portf. w/ highest sharpe ratio,

w/ weights $w_T = \dfrac{\Sigma^{-1}(\mu - R_f \mathbb{1})}{\mathbb{1}^T \Sigma^{-1}(\mu - R_f \mathbb{1})}$

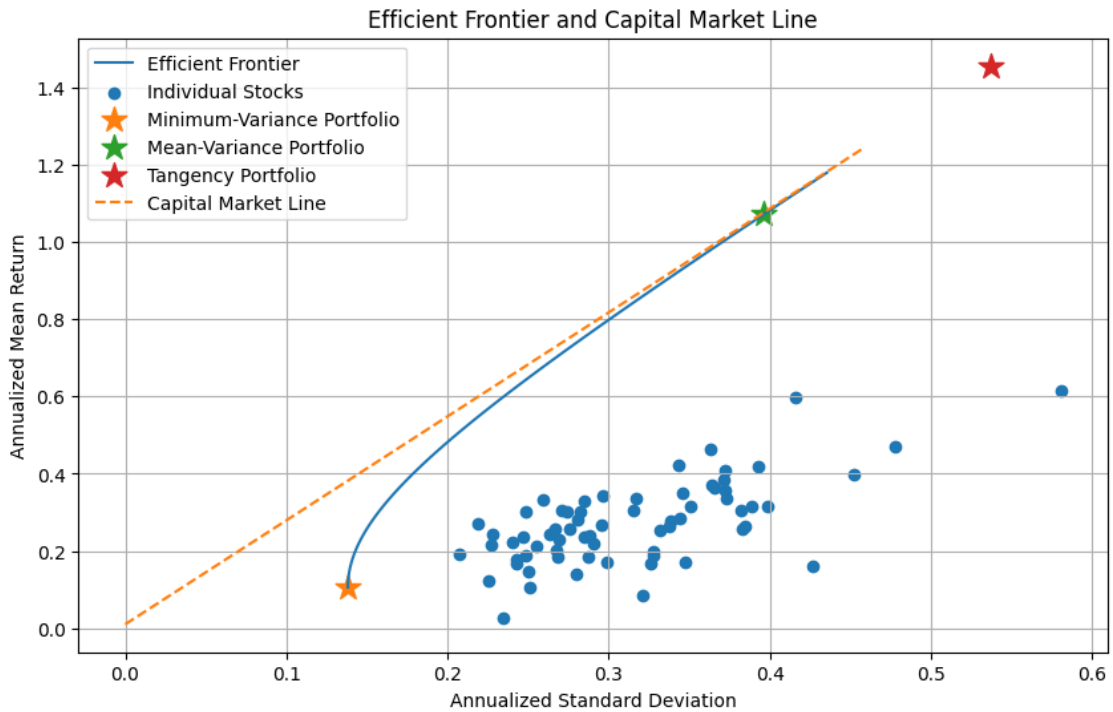
capital market line $\mu_p = R_f + \dfrac{\mu_T - R_f}{\sigma_T}\sigma_p$

[29]:
```
Rf = 0.01

excess_mu = mu_vec - Rf * ones

z = np.linalg.solve(Sigma_mat, excess_mu)
w_tan = z / (ones @ z)
```

```python
tan_return = mu_vec @ w_tan
tan_std = np.sqrt(w_tan @ Sigma_mat @ w_tan)
tan_sharpe = (tan_return - Rf) / tan_std

tan_weights = pd.DataFrame({
    "ticker": tickers,
    "weight": w_tan
})

print("Part (d): Tangency portfolio")
print(f"Annualized mean return: {tan_return:.6f}")
print(f"Annualized standard deviation: {tan_std:.6f}")
print(f"Sharpe ratio: {tan_sharpe:.6f}")

display(tan_weights)

# Capital market line
cml_stds = np.linspace(0, max(frontier_stds) * 1.05, 100)
cml_returns = Rf + tan_sharpe * cml_stds

plt.figure(figsize=(10, 6))

plt.plot(frontier_stds, frontier_returns, label="Efficient Frontier")
plt.scatter(stock_stds, stock_returns, label="Individual Stocks")

plt.scatter(
    minvar_std,
    minvar_return,
    marker="*",
    s=200,
    label="Minimum-Variance Portfolio"
)

plt.scatter(
    mv_std,
    mv_return,
    marker="*",
    s=200,
    label="Mean-Variance Portfolio"
)

plt.scatter(
    tan_std,
    tan_return,
    marker="*",
    s=200,
```

```
    label="Tangency Portfolio"
)

plt.plot(cml_stds, cml_returns, linestyle="--", label="Capital Market Line")

plt.xlabel("Annualized Standard Deviation")
plt.ylabel("Annualized Mean Return")
plt.title("Efficient Frontier and Capital Market Line")
plt.legend()
plt.grid(True)
plt.show()
```

```
Part (d): Tangency portfolio
Annualized mean return: 1.455850
Annualized standard deviation: 0.537046
Sharpe ratio: 2.692230

   ticker    weight
0    AAPL  0.127876
1     ACN  0.464453
2    ADBE  0.211473
3     ADI -0.455824
4    ADSK -0.105321
..    …       …
62     UI  0.223457
63   VRSN -0.059968
64   WDAY -0.322404
65    WDC -0.434434
66   ZBRA  0.209273

[67 rows x 2 columns]
```



[30]:
```
display(Image("3e.png", width=600))
```

e) for each stock, CAPM beta is $\beta_i = \dfrac{\mathrm{cov}(r_i, r_M)}{\mathrm{var}(r_M)}$

return of tangent portf. from pt d)

[32]:
```
market_returns = returns.values @ w_tan

betas = {}

for j, ticker in enumerate(tickers):
    stock_returns_j = returns.iloc[:, j].values

    cov_im = np.cov(stock_returns_j, market_returns, ddof=1)[0, 1]
    var_m = np.var(market_returns, ddof=1)

    betas[ticker] = cov_im / var_m
```

```
beta_series = pd.Series(betas).sort_index()

print("First five ticker-beta pairs:")
display(beta_series.head(5))

print("Last five ticker-beta pairs:")
display(beta_series.tail(5))
```

```
First five ticker-beta pairs:

AAPL    0.201421
ACN     0.161923
ADBE    0.230665
ADI     0.145135
ADSK    0.190252
dtype: float64

Last five ticker-beta pairs:

UI      0.318949
VRSN    0.156395
WDAY    0.170000
WDC     0.103977
ZBRA    0.245606
dtype: float64
```

[33]:
```
display(Image("3f.png", width=600))
```

f) for each stock, regress stock's daily excess return on the market portf. daily excess.

$\rightarrow \quad r_i - R_f = \alpha_i + \beta_i (r_M - R_f) + \varepsilon_i$

The estimated OLS slope coeffs agree w/ the beta vals from pt e) because the slope is $\tilde{\beta}_i = \frac{\mathrm{cov}(r_i, r_M)}{\mathrm{var}(r_M)}$ which is the same formula used in e)!

[35]:
```
Rf_daily = Rf / 252

market_excess = market_returns - Rf_daily
```

```python
ols_betas = {}
ols_alphas = {}

# Regression matrix with intercept
X = np.column_stack([
    np.ones(len(market_excess)),
    market_excess
])

for j, ticker in enumerate(tickers):
    stock_excess = returns.iloc[:, j].values - Rf_daily

    alpha_hat, beta_hat = np.linalg.lstsq(X, stock_excess, rcond=None)[0]

    ols_alphas[ticker] = alpha_hat
    ols_betas[ticker] = beta_hat

ols_beta_series = pd.Series(ols_betas).sort_index()

comparison = pd.DataFrame({
    "CAPM beta": beta_series,
    "OLS beta": ols_beta_series,
    "difference": ols_beta_series - beta_series
})

display(comparison)

print("Maximum absolute difference:")
print(comparison["difference"].abs().max())
```

```
      CAPM beta  OLS beta    difference
AAPL   0.201421  0.201421  5.551115e-17
ACN    0.161923  0.161923  8.326673e-17
ADBE   0.230665  0.230665  1.110223e-16
ADI    0.145135  0.145135  1.665335e-16
ADSK   0.190252  0.190252  0.000000e+00
…          …        …            …
UI     0.318949  0.318949  2.220446e-16
VRSN   0.156395  0.156395  2.775558e-17
WDAY   0.170000  0.170000  8.326673e-17
WDC    0.103977  0.103977  9.714451e-17
ZBRA   0.245606  0.245606  2.775558e-16

[67 rows x 3 columns]

Maximum absolute difference:
2.7755575615628914e-16
```

[ ]:

```

```